

Aitana Morote Martínez

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Research Interests

Natural Language Processing, Mitigating Bias in AI, Quantum Computing

My first research work has focused on the detection and mitigation of biases in language models. More specifically, my work is oriented at reducing the gap in gender bias mitigation in word embeddings between English and morphologically rich languages as Spanish.

In addition, I recently started working on the research line of quantum computing, which is gaining widespread attention as quantum infrastructure is becoming available for scientific research. This technology has a great potential in many fields, such as Natural Language Processing. I particularly focused on the formulation of Instance Selection (IS) and clustering problems as Quadratic Unconstrained Binary Optimization (QUBO) problems.

Education

University of Alicante, Alicante, Spain
M.S. in Data Science

Sept 2025 - Sept 2027

University of Alicante, Alicante, Spain
B.S. in Computer Engineering

Sept 2019 - July 2024

- GPA: 8.45/10
- Graduation project: The problem of grammatical gender in Spanish to mitigate gender bias and stereotypes. (Advisor: Elena Lloret)

Sungkyunkwan University, Suwon, South Korea
Global Mobility Program

Feb 2022 - July 2022

Professional Experience

**Digital Intelligence Center (CENID),
Language and Information Systems Group (GPLSI)**

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Nov 2024 - present

- Collaborated on the NEL-VIVES project (Language Technology Plan for Valencian language)
- Collaborated on the CORTEX project (COncscious natuRAL TEXt generation)
- Collaborated on the ALIA project (The public AI infrastructure in Spanish and co-official languages)

NTT DATA (internship)

Feb 2023 - July 2023

- Performed R&D tasks using Cloud Computing technologies.

Applied Intelligent Architectures (AIA) research group

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Sep 2022 - Dec 2022

- Studied the state-of-the-art in the research line on Computer Vision and AI applications for the early detection of dementia and prevention of Alzheimer's Disease.

Publications

Preserving Grammatical Gender when Debiasing Word Embeddings in Spanish 

2025

Aitana Morote Martínez, Juan Pablo Consuegra-Ayala, Elena Lloret

- Issue No. 75 of the journal and for presentation at the XLI International SEPLN Conference.

GPLSICORTEX at SemEval-2025 Task 10: Leveraging Intentions for Generating Narrative Extractions 

2025

Iván Martínez-Murillo, María Miró Maestre, **Aitana Morote Martínez**, Snorre Ralund, Elena Lloret, Paloma Moreda, Armando Suárez Cueto

- In Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025) (pp. 575–583). Association for Computational Linguistics.

Team GPLSI at QCléf 2025: Quantum-Inspired Instance Selection and Clustering [🔗](#)

2025

Juan Pablo Consuegra-Ayala, **Aitana Morote Martínez**, Francisco Valero-Abellón, Elena Lloret, Paloma Moreda, Manuel Palomar

- In Working Notes of CLEF 2025 – Conference and Labs of the Evaluation Forum. Madrid, Spain: CEUR Workshop Proceedings.

Projects

NEL-VIVES (Language technology plan for Valencian language)

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Funding entity: Ministry of Economic Affairs and Digital Transformation, European Union NextGenerationEU/PRTR

Nov 2024 – May 2025

- Designed a new method for reducing gender bias in Spanish word embeddings (*INLP-Gram* [🔗](#)).
- Created a dataset of Spanish words to aid the evaluation and improvement of gender bias mitigation techniques in Spanish.
- Provided a detailed evaluation of the proposed method, showing its effectiveness and exploring how it handles grammatical and societal gender information.

CORTEX (COnscious natuRal TEXt generation)

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Funding entity: MCIN/ AEI/10.13039/501100011033/ and “FEDER Una manera de hacer Europa”

Nov 2024 – May 2025

- Participated as part of the team GPLSICORTEX in the SemEval-2025 Task 10 subtask 3:
 - Studied the effect of the identification of propaganda strategies on a narrative extraction task.
 - Collaborated on the elaboration of a a curated dataset with labeled communicative intentions for the same task.
- Collaborated in a longitudinal Large Language Models Analysis:
 - Used prompt engineering techniques to assess critical gaps currently affecting LLMs.
 - Performed a rigorous, robust, and consistent qualitative evaluation of outputs across all gaps, tasks, and models.
 - Examined the trade-offs between the potentials and limitations of current LLMs across a diverse set of tasks.
- Participated as part of team GPLSI at Quantum Cléf 2025:
 - Designed a method to solve an instance selection problem (i.e., finding the more relevant instances to train a language model) using Local Sets, which is formulated using a QUBO function in order to be executed on a Quantum Annealer.

ALIA (The public AI infrastructure in Spanish and co-official languages)

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Funding entity: European Union NextGenerationEU/PRTR

June 2025 – present

- Developed a support platform to guide the use of ALIA family models (foundational language models (LLMs) in Spanish and co-official languages) and datasets: *ALIA support* [🔗](#).
- Created manuals and tutorials to teach how to use the models.

Competitions and challenges

Quantum CLEF (QCléf) Lab at CLEF 2025

Apr 2025 – present

- Participation as part of the team GPLSI in the Task 2—Instance Selection— and Task 3—clustering— of the Quantum CLEF (QCléf) Lab at CLEF 2025, which explores the integration of quantum computing technologies into real-world information retrieval and machine learning workflows.

- In Task 2, our team achieved the best trade-off between effectiveness and reduction.

SemEval-2025 Task 10: Leveraging Intentions for Generating Narrative Extractions

Oct 2025 – Feb 2025

Ranked 6th position

- Participation as part of the GPLSI team in the SemEval-2025 Task 10 subtask 3 for the English language, which is focused on narrative extraction given news articles with a dominant narrative. We design an external knowledge injection approach to fine-tune a Flan-T5 model so the generated narrative explanations are for the provided dominant narrative in each text.

Innovation Banking Hack Fest (IBHF) 25

17, 18, 19 Oct 2025

- 1st Prize to the best technical solution in the IBHF 25 hackathon with the objective of designing and developing an innovative application in the banking environment for the purpose of offering and improving the customer experience in digital banking.

Innovation Banking Hack Fest (IBHF) 23

27, 28, 29 Oct 2023

- Participation in the IBHF 23 hackathon with the objective of designing and developing an innovative application in the banking environment for the purpose of offering and improving the customer experience in digital banking.

Talks

Evaluation and Biases in NLP: Preserving Grammatical Gender when Debiasing Word Embeddings in Spanish

14 Sept 2025

TecnoLing 2025, XLI Congreso de la SEPLN, Instituto Tecnológico de Aragón

Mitigation of gender bias in word embeddings in Spanish

8 Nov 2024

Digital Intelligence Center (CENID)

Languages

Languages: Spanish (native), French (fluent), English (fluent), Korean (beginner)

Programming languages that I have learned and used: Python, C++, C, Java, C#, SQL, R